Moral Tensors and DecisionProofs

Compiling Language into an Auditable, Grounded Safety Layer

Structure-preserving representation before decision contraction.

`pip install erisml-compiler` · `github.com/ahb-sjsu/erisml-compiler · erisml-lib · agi-hpc`

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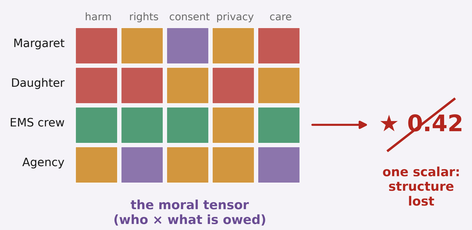
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PyTorch Conference North America 2026 — Responsible AI poster

1

Don't collapse to a scalar.



✓ keep the structure

who is affected
what is owed
who consented
who bears imposed risk

contract to a decision only at the end, and log the contraction: weights · who lost · what residue remains

2

pip install erisml-compiler — structure before the scalar



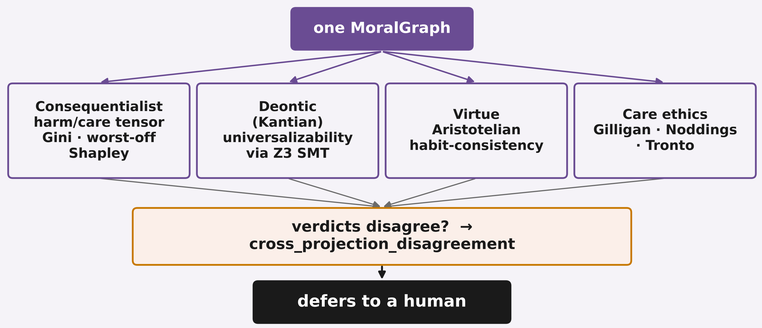
MoralGraph nodes: stakeholder · act · maxim · commitment · fact · norm carries a canonical SHA-256 hash

extractor tiers: rule (deterministic, silicon-castable)
LLM (NRP / vLLM + critic) · probe (LaBSE head)

alpha v0.9.0 · 477 tests · MIT · Zenodo DOI 10.5281/zenodo.20659432

3

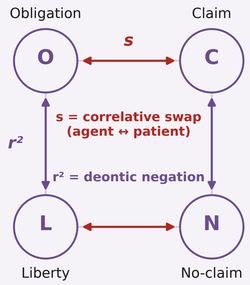
Four lenses. No silent winner.



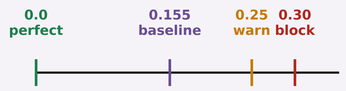
"We run four and show you where they conflict — we don't pick one and hide it."

4

Is this 'good'? → Does it preserve the bonds?



s and r² commute → V4
(Klein four-group, order 4): measured



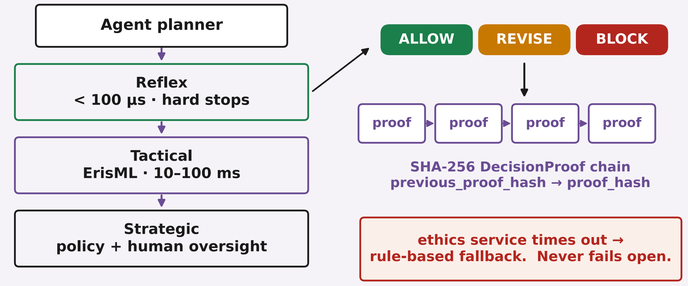
Bond Index: does the judgment survive the swap?

V4 machine-checked:
Lean 4 + Mathlib

full D4 (order 8) posited, now tested: a 09/26 hunt in a small LM found no quarter-turn. V4 stands.

5

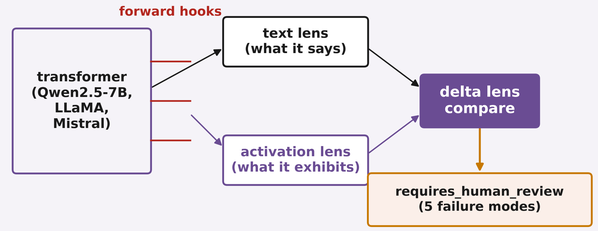
Safety in the loop. Fails safe.



between the agent's planner and its actuators

6

What the model says vs what it exhibits



Activation lens: first calibration (09/26). Consumer-metric gating catches 19-21 pp more real behavior changes than raw L2 at matched flag rates. Replicated 2/2: second model + second transform. Probe mid-layers: the final-layer β̂ map fails held-out validation in all three runs.

turboquant-pro (PyPI):
PyTorch-native compression
HF 1-liner: TurboQuantCache()
Triton + Volta kernels · vLLM ~5× KV
live here: 3-bit codec on NATS
claims CI-gated · false-clear diag (v2.0.0a3)

7

Replay the judgment, and the regime that made it

full_response	NORMAL	EMERGENCY
Margaret	**0.185 forbid**	**0.705 prefer**
Daughter	**0.402 neutral**	**0.902 prefer**
EMS crew	**0.318 forbid**	**0.902 prefer**
Agency	**0.200 forbid**	**0.802 prefer**

DEME's scores: stand_by 0.834 → full_response 0.804, and stand_by inverts to 0.307

elevation: detector 0.97 ≥ 0.90 → GRANTED, paramedics only, auto-reverts, both edges audited

conf 0.89 → REFUSED: human review and the EMS call, no record share

worst-off: Daughter → Margaret
Shapley authority: Margaret → EMS

Three DecisionProofs: normal, emergency, the elevation edge

$ python -m erisml.examples.care_robot_regimes reproduces every number

In the robot, DEME is a firewall: it vets each proposed act, never picks one

proposed act → **governor attested sensors** → **constitution signed rules** → **DEME can veto** → **signed manifest + body lock** → **act**

any stop: robot holds

Change one input, nothing else:	base	one changed
fire call: one of two detectors silent	**passed**	**stopped: governor**
ambulance: her refusal heard / not	**stopped: DEME**	**passed**
help her up: she asked / did not	**passed**	**stopped: DEME**

9 pairs, each prediction committed before its run; 2 of 18 wrong, both explained in the record

Found by this test, then fixed:
the lift she had not asked for passed DEME until 18 Oct.
Rerun: only that verdict changed.

Simulated 3D home, scripted proposals, no hardware

1. Structure before contraction — keep the tensor, log the collapse + the residue.
2. Pluralism is the responsible move — four lenses, defer to a human on conflict.
3. Auditable by construction; fails safe, never open.

alpha v0.9.0 · text path solid · activation lens early; first calibration 09/26 (consumer-metric, replicated 2/2) · ranks 1–3 real, higher partial
V4 measured, D4 posited (Lean + Mathlib verified)
embodied = simulated 3D home, scripted, no hardware · agents = live

companion paper · erisml-compiler · erisml-lib · agi-hpc